Nicholas Joseph Emile Richardson

Curriculum Vitae

Hamilton Hall, Room 103E
McMaster University
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njericha.github.io

Research Interests

Clustering, Audio/Signal Processing, Tensor Factorization, and Continuous Optimization

Education

Sep. 2022 – Jul. 2026 **Doctor of Philosophy**
University of British Columbia, Vancouver, BC
Institute of Applied Mathematics, supervised by Michael P. Friedlander
Thesis: *Computational Techniques for Analysis of Mixed Signals*

Sep. 2020 – May 2022 **Master of Mathematics**
University of Waterloo, Waterloo, ON
Applied Mathematics Department, supervised by Giang Tran
Thesis: *A Sparse Random Feature Model for Signal Decomposition*

Sep. 2016 – Apr. 2020 **Bachelor of Mathematics**
University of Waterloo, Waterloo, ON
Mathematical Physics Major, Pure Mathematics Minor

Publications

2025 Joel Saylor, **Nicholas Richardson**, Naomi Graham, Robert Lee, and Michael P. Friedlander. Tracking cu-fertile sediment sources via multivariate petrochronological mixture modelling of detrital zircons. *Journal of Geophysical Research: Earth Surface*, 130, e2025JF008406.

2025 Naomi Graham, **Nicholas Richardson**, Michael P. Friedlander, and Joel Saylor. "Tracing Sedimentary Origins in Multivariate Geochronology via Constrained Tensor Factorization. *Mathematical Geosciences*, 57, 601–628.

2023 Michael L. Waite and **Nicholas Richardson**. Potential Vorticity Generation in Breaking Gravity Waves. *Atmosphere*, 14(5):881.

2023 **Nicholas Richardson**, Hayden Schaeffer, and Giang Tran. SRMD: Sparse Random Mode Decomposition. *Communications on Applied Mathematics and Computation*, 6, 879–906.

2023 Lam Si Tung Ho, **Nicholas Richardson**, and Giang Tran. Adaptive group Lasso neural network models for functions of few variables and time-dependent data. *Sampling Theory, Signal Processing, and Data Analysis*, 21(2):28.

Submitted

2026 **Nicholas Richardson**, Noah Marusenko, and Michael P. Friedlander. Multiple Scale Methods for Optimization of Discretized Continuous Functions. arxiv.org/abs/2512.13993.

Experience & Teaching

Sep. 2026 – Present **Postdoc / Instructor**
McMaster University, Hamilton, ON
Supervised by Paul McNicholas
- Applied Regression Analysis with SAS (Fall 2026)

Sep. 2023 – Apr. 2026 **Instructor**
University of British Columbia, Vancouver, BC
- Intro to Computing: Python (Fall 2025, Winter 2026)
- Calculus I (Fall 2023, Winter 2025)

Sep. 2022 – Apr. 2024 **Teaching Assistant**
University of British Columbia, Vancouver, BC
- Calculus I (Fall 2024)
- Calculus II (Winter 2023)
- Calculus III (Fall 2022)

Sep. 2022 – Apr. 2024 **Mathematics Learning Centre Tutor**
University of British Columbia, Vancouver, BC

Jun. 2020 – Dec. 2024 **Mathematics and Physics Tutor**
Self-Employed, Remote
High School and Undergraduate Levels

Jan. 2018 – Dec. 2021 **Teaching Assistant**
University of Waterloo, Waterloo, ON
- Mathematics of Music (Fall 2020, 2021)
- Special Topics in Mathematical Connections: Math & Music (Fall 2021)
- Complex Analysis (Winter 2020)
- Calculus 1 & 2 (Fall 2019, Winter 2018, 2020, 2021)

Awards

McMaster University

Sep. 2026 Britton Postdoctoral Fellowship

Society of Industrial and Applied Mathematics

Jun. 2026 OP26 Student Travel Award

Jul. 2024 AN24 Student Travel Award

Natural Sciences and Engineering Research Council of Canada

May 2023 Canadian Graduate Scholarship — Doctoral (330 awards nationwide)

May 2021 Canadian Graduate Scholarship — Master's (840 awards nationwide)

May 2019 Undergraduate Student Research Award (3 149 awards nationwide)

University of British Columbia

May 2026 Graduate Student Travel Award

May 2022 Faculty of Science PhD Tuition Award

Sep. 2022 Four Year Doctoral Fellowship

Sep. 2022 President's Academic Excellence Initiative PhD Award

University of Waterloo

May 2021 President's Academic Excellence Initiative PhD Award

Aug. 2020 K.D. Fryer Gold Medal

Jun. 2020 Dean's Honours List With Distinction

May 2019 President's Research Award

Jan. 2019 Paul Berg Memorial Award

Jan. 2019 Arthur Beaumont Memorial Scholarship

Sep. 2016 President's Scholarship of Distinction

Sep. 2016 Alumni @ Microsoft Entrance Scholarship in Mathematics

Presentations & Posters

Jun. 2026 **SIAM Conference on Optimization**
University of Edinburgh, Edinburgh, UK
Multiscale Methods for Optimization of Discretized Continuous Functions (Oral)

Jan. 2026 **Operation Research Seminar**
Simon Fraser University, Burnaby, BC
Multiscale Methods for Optimization of Discretized Continuous Functions (Oral)

Dec. 2025 **Geology Industry Training**
Teck and BHP, Vancouver, BC
Tucker-1 Demixing Model: Under the Hood (Oral)

Jul. 2025 **Joint SIAM/CAIMS Annual Meetings**
Montréal Convention Centre, Montréal, QC
An Optimization Framework for Constrained Tensor Factorization (Oral)

Jul. 2025 **CAIDA/TrustML Workshop — ICML Visitors @ UBC**
University of British Columbia, Vancouver, BC
Unsupervised Signal Demixing with Tensor Factorization (Poster)

Jul. 2025 **Vision and Learning Workshop @ ICML**
Simon Fraser University, Vancouver, BC
Unsupervised Signal Demixing with Tensor Factorization (Poster)

Dec. 2024 **Canadian Mathematical Society Winter Meeting**
Kwantlen Polytechnic University, Richmond, BC
Density Separation with Tensor Factorization (Oral)

Oct. 2024 **INFORMS Annual Meeting**
Seattle Convention Center, Seattle, WA
Density Separation with Tensor Factorization (Oral)

Sep. 2024 **West Coast Optimization Meeting**
University of British Columbia, Vancouver, BC
Density Separation with Tensor Factorization (Oral)

Oct. 2023 **Biennial Meeting of SIAM Pacific Northwest Section**
Western Washington University, Bellingham, WA
Non-negative Tensor Decompositions for Unsupervised Learning (Oral)

Jun. 2021 **Canadian Applied and Industrial Mathematics Society**
University of Waterloo, Waterloo, ON
Sparse Random Wavelet Representation & Decomposition (Poster)
Runner up in annual poster competition

Software

Published Code & Packages

2026 **UBC Thesis Typst Template**, *Typst*
github.com/njericha/ubc-thesis-typst-template

2024 – 25 **MatrixTensorFactor.jl & BlockTensorFactorization.jl**, *Julia*
github.com/MPF-Optimization-Laboratory/BlockTensorFactorization.jl

2023 **SedimentSourceAnalysis.jl**, *Julia*
github.com/njericha/SedimentSourceAnalysis.jl

2022 **Sparse Random Mode Decomposition**, *Python*
github.com/GiangTTran/SparseRandomModeDecomposition

2017 **Drasil**, *Haskell*
jacquescarette.github.io/Drasil

Computer Skills

Julia, Python, Typst, MATLAB, Quarto, LaTeX, Adobe Software, Haskell

Journal Papers Reviewed

2025 SIAM Journal on Scientific Computing (1 review)

2023 – 26 Springer's Journal of Optimization Theory and Applications (3 reviews)

Community & Leadership

Sep. 2025 – Aug. 2026 **SIAM Student Chapter**
University of British Columbia, Vancouver, BC
- Co-organizer for the first UBC Applied Mathematics Meeting (Oct. 2025)

Sep. 2023 – Aug. 2026 **Math Graduate Committee**
University of British Columbia, Vancouver, BC
- Co-organizer and Seminar Host (2024–2026)
- Social Convenor (2023–2025)

Student Supervision

May 2026 – Aug. 2026 **Mehdi Naami**
University of British Columbia, Vancouver, BC
Undergrad Research Project: Factorization with Automatic Differentiation

May 2026 – Aug. 2026 **Michael Yang**
University of British Columbia, Vancouver, BC
Master's Student Summer Project: Moment Matching for Density Estimation.

Sep. 2024 – Jan. 2025 **Noah Marusenko**
University of British Columbia, Vancouver, BC
Undergrad Research Project: Multiscale Methods for Tensor Factorizations

Additional Research

May 2019 – Aug. 2019 **Computational Fluids Research Assistant**
University of Waterloo, Waterloo, ON
Supervised by Michael L. Waite
Project: Vorticity generation simulations with MATLAB and FORTRAN

May 2017 – Aug. 2017 **Software Engineering Research Assistant**
McMaster University, Hamilton, ON
Supervised by Spencer Smith and Daniel Szymczak
Project: Automatic scientific code & documentation generation with Haskell

Last updated: 01 September 2026